

SOCKETS

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1. Server_side.c

```
rehan_lenovo@Rehan-Lenovo x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int server_fd, client_fd;
    struct sockaddr_un address;
    char buffer[100];

    server_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    unlink(SOCKET_PATH);

    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    listen(server_fd, 5);

    printf("Server waiting...\n");

    client_fd = accept(server_fd, NULL, NULL);

    read(client_fd, buffer, sizeof(buffer));

    printf("Message from client: %s\n", buffer);

    close(client_fd);
    close(server_fd);

    unlink(SOCKET_PATH);

    return 0;
}
```

2. Client_side.c

```
rehan_lenovo@Rehan-Lenovo x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int client_fd;
    struct sockaddr_un address;

    client_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    connect(client_fd,
            (struct sockaddr *)&address,
            sizeof(address));

    write(client_fd, "Hello Server", 12);

    printf("Message sent to server.\n");

    close(client_fd);

    return 0;
}
```

Output:

```
rehan_lenovo@Rehan-Lenovo x + v
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc server_side.c -o server_side
rehan_lenovo@Rehan-Lenovo:~/OS$ ./server_side
Server waiting...
Message from client: Hello Server
rehan_lenovo@Rehan-Lenovo:~/OS$

rehan_lenovo@Rehan-Lenovo x + v
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc client_side.c -o client_side
rehan_lenovo@Rehan-Lenovo:~/OS$ ./client_side
Message sent to server.
rehan_lenovo@Rehan-Lenovo:~/OS$
```